

INFLECTION

Real-time multilingual sentiment analysis

MSML 641 - Mid Semester Presentation

Sawyer Byrd, Andrew Liu, Alessandro Vivaldi



“A real-time affordable call companion that transcribes, translates, and reads customer sentiment as you talk.”

The Team:

- Sawyer Byrd - Researcher, Developer, Software testing
- Andrew Liu - Design planner, Software testing, Editing
- Alessandro Vivaldi - Researcher, Developer, Editing

PERSONA

A QA manager or team lead at a BPO or international support organization:

- Handles calls where customers may not be fully fluent
- Can prompt customers to explain in their native language
- Needs real-time emotional signal, not post-call transcript review

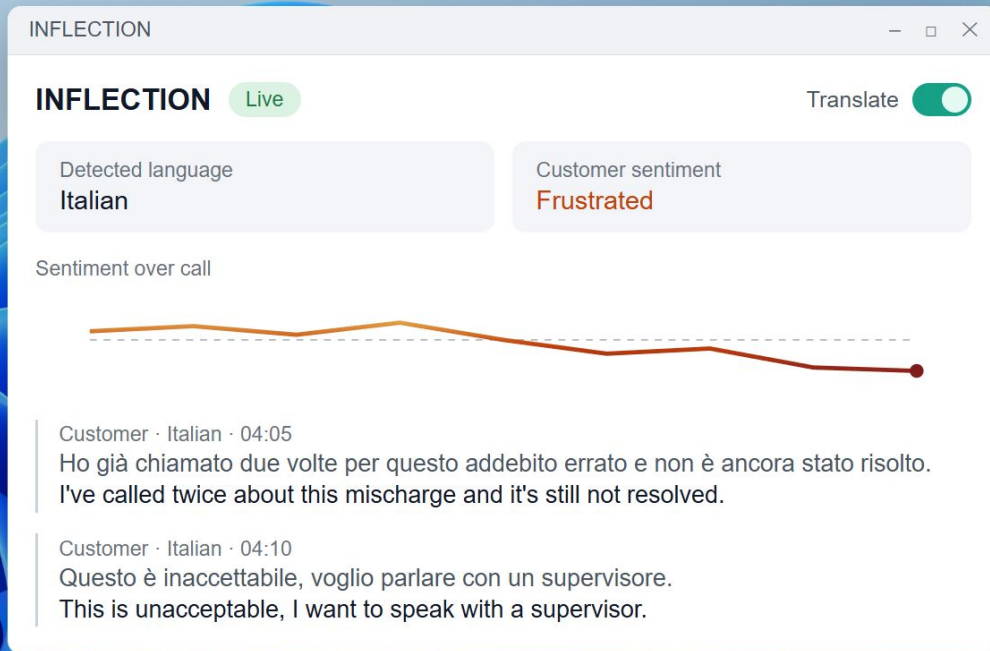
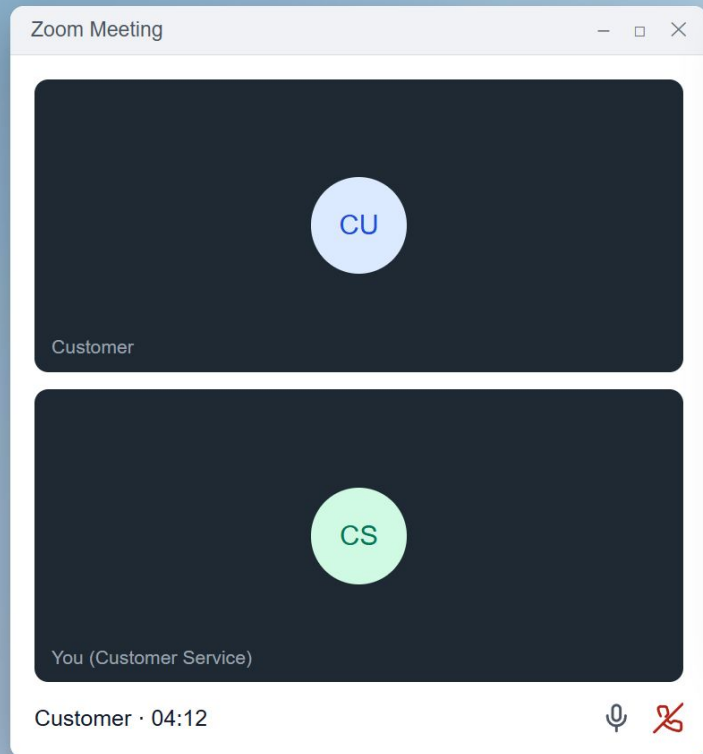
PAIN

Agents can handle the conversation but not the emotional nuance lost across language barriers, and existing solutions are expensive:

- Enterprise tools (SupportLogic, CallMiner) out of reach for small/mid-size BPOs
- Existing tools marketed as 'voice sentiment' analyze transcripts, not acoustic tone (emotional signal loss)
- No accessible tool combines text + tone analysis

Demo

ENVISIONED END PRODUCT



USER VALIDATION

Testing

- Languages tested: Italian, Japanese on native speakers
- Session format: length, structure (5 min guided conversations per emotion state)
- Environment: terminal-based, no UI
- Method: moderated by developers

Strengths

- Few, clear instructions to follow once the program is running
- Realtime translation/transcript of the chat, and correct identification of language to translate from
- Tone analysis could distinguish complex emotional differences

Weaknesses

- How to set up the actual tool (missing UI)
- Tone-based sentiment analysis would sometimes fail to classify high intensity emotions in specific circumstances.
- Translator misses

Findings

- A mix of text and tone sentiment analysis may help the classifier more.
- Automatic language detection kept the tool intuitive.
- Sentiment analysis input needs to be consistent with translator/transcription input
- Reusing one constructor per language (latency)

NORTH-STAR

Emotional Escalation Flagging

**Combining Voice Tone + Text
Sentiment**

Goal is Accurate real-time
emotional escalation flagging

CURRENT STATE

3 of 5

Components Complete

Core pipeline fully assembled.
Running in parallel:

- Live transcription
- Multilingual translation
- Acoustic tone sentiment

NEXT STEPS

Remaining Tasks

- Text sentiment layer
- Combined flagging logic
- UI & integration

*Pipeline went from zero to
end-to-end functionality these
past couple of weeks. Accuracy
testing begins once layers
integrate.*